

Dexter Fernandes

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Robotics Software Engineer

Robotics Software Engineer with 3+ years of experience building low-latency perception and edge AI systems in fast-moving start-up environments. Experienced across C++17, Python, ROS2, point cloud processing, computer vision, and hardware-constrained inference, with a track record of taking systems from data and model development through optimisation, deployment, and field validation. MSc Robotics graduate with hands-on work in SLAM, localisation, sensor integration, and research-led robotics engineering. Comfortable working across research and engineering, and well suited to scrappy teams where rapid iteration and real-world robustness matter.

WORK EXPERIENCE

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Owned the end-to-end computer vision pipeline for real-time perception systems in harsh industrial environments, from data curation and model training through deployment and on-site validation, improving F1 by 15% since project takeover while maintaining deterministic 24/7 operation.
- Built multimodal data ingestion, cleansing, training, and validation pipelines using Python, Pandas, Apache Airflow, MLflow, DVC, Optuna, and CVAT, cutting model iteration time by 40% and reducing rollout time from 3 days to under 24 hours.
- Refactored legacy services into Dockerised microservices with Kafka-based messaging, structured logging, health checks, retries, and safe restarts, improving fault isolation, increasing MTBF by 35%, and reducing live video pipeline crashes by 67%.
- Benchmarked and optimised edge inference workflows across latency, memory, and stability constraints, defining repeatable FP16 validation paths and acceptance thresholds for reliable deployment across hardware targets.
- Led field validation and root-cause analysis on BP oil-rig deployments, diagnosing failures under environmental stressors and translating performance findings into practical fixes that improved robustness and reduced false positives in live operation.

LivNSense Technologies • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI • Full-time

- Led a 4-member team to design, train, and deploy production object detection and tracking models for industrial safety applications, reducing false positives by 33% and achieving sub-50 ms inference latency on edge devices.
- Built low-latency perception pipelines using OpenVINO, and C++17, increasing throughput from 5 FPS to 15 FPS and delivering deterministic real-time inference on Intel VPU hardware under tight power and memory constraints.
- Defined optimisation workflows spanning quantisation-aware training, pruning, layer fusion, FP16/INT8 deployment, and profiling, reducing model size by 50% while meeting accuracy, latency, and memory targets.
- Built automated data, labelling, and performance-tracking pipelines, repurposing older models for annotation and cutting manual labelling effort by 80% and also developed synthetic-data tooling to improve edge-case coverage and reduce manual labelling time by 60%.
- Worked closely with hardware, software, and QA teams in a fast-moving start-up environment to integrate perception modules into deployed systems, improve delivery consistency, and guide experimentation and optimisation for production releases.

PROJECTS

ROS2 SLAM & Nav2 Simulation

Feb 2026

- Built a simulation-based robotics stack in ROS2 Jazzy and Gazebo Harmonic to evaluate LiDAR-centric SLAM, LiDAR+RGB fusion SLAM, and autonomous navigation in structured scenarios.

- Benchmarked RTAB-Map, LIO-SAM, Cartographer, and GLIM across 2D and 3D LiDAR configurations, comparing mapping and localisation behaviour to support system selection and iteration.
- Developed ROS2 nodes, launch files, and parameterised configurations in C++ and Python, using tf2 and message synchronisation to maintain frame consistency and debug transform timing issues.
- Implemented point cloud processing with PCL and Open3D for filtering, segmentation, and registration, and added an EKF-based localisation layer with robot_localization to fuse wheel odometry and IMU for smoother odom-to-base tracking during LiDAR-only SLAM.
- Containerised the stack with Docker on Linux and managed versioning in Git to support reproducible builds and consistent execution across development environments.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Radiological SLAM with LiDAR Odometry. Designed and implemented a real-time SLAM system using RTAB-Map on the Clearpath Husky platform, integrating LiDAR and radiological data for localisation and mapping in inspection scenarios.
- Evaluated RTAB-Map, LIO-SAM, and Cartographer across comparable scenarios to assess mapping and localisation behaviour in simulation before real-world deployment.
- Validated sensor placement for coverage and calibrated LiDAR extrinsics to the robot base, including the Velodyne VLP-16 and Intel RealSense D435i.
- Implemented a ROS2 (rclcpp) point cloud pipeline in C++17 using PCL for filtering, segmentation, and registration.
- Developed and maintained ROS2 nodes using tf2 and message synchronisation, and debugged tf trees and frame connectivity.

Coursework:

- Hands-on work with inverse kinematics and closed-loop control on a laboratory robotic arm platform.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, perception systems, and robotics-adjacent engineering.

SKILLS

- **Robotics Stack:** Closed-loop control, Closed-loop robot control, Gazebo (Harmonic), Inverse kinematics, Localisation, Nav2, rclcpp, rclpy, Real-time perception systems, ROS2 (Robot Operating System 2), Sensor integration, SLAM, tf2
- **SLAM & 3 Perception:** Cartographer, Ceres, Eigen, Filtering, GLIM, LiDAR SLAM, LIO-SAM, Point cloud processing, Pose graph optimisation, Registration, RTAB-Map, Scan matching, Segmentation
- **Programming & Tooling:** Bash, C++17, CMake, colcon, Git, Pandas, Python
- **Computer Vision & Image Processing:** Data augmentation, Image processing, LiDAR extrinsics calibration, Object detection, Open3D, OpenCV, PCL (Point Cloud Library), Sensor calibration
- **Deep Learning and Optimisation:** Bayesian Optimisation, Layer fusion, ONNX, OpenVINO, Optuna, Post-training Quantisation, PyTorch, Quantisation-Aware Training, Structured pruning, TensorFlow, TensorRT
- **Cloud, Edge & MLOps:** Apache Airflow, Apache Kafka, Docker, DVC (Data Version Control), MLFlow
- **AI-Assisted Development:** ChatGPT Codex, Claude Code, Pompt engineering